

Game Plaza - Project Summary & Development Plan

PROJECT OVERVIEW

****Game Plaza**** is a gamer social platform MVP that allows users to:

- Create profiles and manage game libraries
- Find other gamers with shared interests
- Form matches and tournaments
- Build social connections through follows/likes

****Technology Stack:****

- ****Backend****: PHP (vanilla, no framework) + MySQL + PDO
- ****Frontend****: HTML5 + Bootstrap 5 + JavaScript (Fetch API)
- ****Animations****: AOS (Animate On Scroll)
- ****Security****: Password hashing, prepared statements, session management

CURRENT STATUS

 ****COMPLETED:****

- [x] Project structure created
- [x] WBS (Work Breakdown Structure) - 60+ tasks organized
- [x] GANTT timeline - 7 phases, ~21-day estimate
- [x] RBS (Risk Breakdown Structure) - 20+ risks identified with mitigation
- [x] Database schema designed - 8 tables, relationships, indexes

DEVELOPMENT APPROACH

"Phase-Based Iterative Development"

We'll build in ****7 distinct phases****, each producing working code:

1. ****PHASE 1 - FOUNDATION**** (This will be next)

- Create config files
- Database connection setup
- Basic helpers/utilities

2. ****PHASE 2 - BACKEND CORE****

- SQL schema import
- Model classes (User, Game, Match, etc.)
- Authentication endpoints

3. ****PHASE 3 - API ENDPOINTS****

- User management
- Game library operations
- Matchmaking logic

4. ****PHASE 4 - FRONTEND PAGES****

- HTML templates
- Bootstrap layout
- Base structure

5. ****PHASE 5 - JAVASCRIPT****

- Fetch API utilities
- Dynamic interactions
- DOM updates

6. ****PHASE 6 - ANIMATIONS & POLISH****

- AOS integration
- Styling refinement
- Responsive tweaks

7. ****PHASE 7 - TESTING & DOCS****

- Manual QA
- Documentation
- Final cleanup

PROJECT STRUCTURE

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game-plaza/
├── config/                                # Configuration files
│   ├── db.php                           # Database connection
│   ├── session.php                       # Session handler
│   └── config.php                        # App constants
├── api/                                  # API endpoints (JSON responses)
│   ├── auth/
│   │   ├── register.php
│   │   ├── login.php
│   │   └── logout.php
│   ├── users/
│   │   ├── get.php
│   │   ├── update.php
│   │   └── search.php
│   ├── games/
│   │   ├── add.php
│   │   ├── remove.php
│   │   ├── library.php
│   │   └── search.php
│   ├── matchmaking/
│   │   ├── suggestions.php
│   │   └── filter.php
│   └── matches/
│       ├── create.php
│       ├── join.php
│       ├── list.php
│       └── details.php

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├── social/
│   ├── like.php
│   ├── follow.php
│   └── followers.php
├── models/                                # Data access layer
│   ├── User.php
│   ├── Game.php
│   ├── UserGame.php
│   ├── Match.php
│   ├── Preference.php
│   └── Database.php
├── views/                                # HTML template pages
│   ├── layout.php                        # Master template
│   ├── index.php                        # Home page
│   ├── login.php                        # Login form
│   ├── register.php                     # Registration form
│   ├── profile.php                      # User profile
│   ├── games.php                        # Game library
│   ├── matchmaking.php                  # Suggestions
│   └── matches.php                      # Tournaments/Matches
├── assets/                               # Static files
│   ├── css/
│   │   └── style.css                    # Main stylesheet
│   ├── js/
│   │   ├── api.js                      # Fetch utilities
│   │   ├── profile.js
│   │   ├── games.js
│   │   ├── matchmaking.js
│   │   └── matches.js
│   └── img/                             # Images/icons
├── docs/                                 # Documentation
│   ├── WBS.md
│   ├── GANTT.md
│   ├── RBS.md
│   ├── DATABASE_SCHEMA.md
│   └── README.md                        # (Will create)
├── index.php                            # Entry point / router
└── game_plaza.sql                       # Database dump (Will create)

```

KEY FEATURES (MVP)

✓ Authentication

- Register new account
- Login with password verification
- Logout & session management
- Password hashing (bcrypt)

✓ User Profiles

- Username, email, bio
- Skill level (Beginner → Pro)
- Availability status
- Follow/Like system

✓ Game Library

- Add games from predefined list
- Assign platform (Steam, Epic, Blizzard, etc.)
- Remove games
- View own library

✓ Matchmaking

- Find users with shared games
- Compatibility score algorithm
- Filter by preferences
- Suggest compatible players

✓ Matches & Tournaments

- Create matches for specific games
- Join/Leave matches
- View participants
- Track match status

✓ Social Interactions

- Follow other players
- Like profiles
- View followers
- Basic user discovery

✓ Frontend

- Responsive design (Bootstrap 5)
- Real-time form interactions
- Animated elements (AOS)
- Error handling & feedback

DATABASE (8 Tables)

Table	Purpose	Records
users	User accounts & profiles	~10 sample
games	Game catalog	~20 sample
user_games	User's library (Many-to-Many)	~50 sample
user_preferences	User preferences & interests	~10 sample
matches	Tournaments/Matches	~5 sample
match_participants	Users in matches	~15 sample
likes_follows	Social connections	~20 sample
compatibility_scores	Cache layer (optional)	Computed

SECURITY MEASURES

✅ ****Implemented:****

- PDO prepared statements (SQL injection prevention)
- password_hash() / password_verify() (bcrypt hashing)
- Session regeneration on login
- CSRF token implementation
- Input validation & sanitization
- htmlspecialchars() for XSS prevention
- SQL error suppression

TESTING CHECKLIST

- [] All CRUD operations working
- [] API endpoints return correct JSON
- [] Authentication flow complete
- [] Matchmaking suggestions accurate
- [] Frontend interactions responsive
- [] No SQL errors in database
- [] Responsive on mobile & desktop
- [] Form validation working
- [] Error messages displayed properly

NEXT STEPS

****READY TO BEGIN?****

When you're ready, I'll create:

****PHASE 1: FOUNDATION****

1. ****config/db.php**** - Database connection handler
2. ****config/config.php**** - App constants & settings
3. ****config/session.php**** - Session configuration
4. ****models/Database.php**** - PDO abstraction layer
5. ****game_plaza.sql**** - Complete database schema with sample data

This will give us a solid foundation to build the API on.

ASSUMPTIONS & NOTES

- ✓ Local MySQL server (XAMPP/Laragon/etc) available
- ✓ PHP 7.4+ installed
- ✓ Bootstrap 5 via CDN
- ✓ No real external APIs (mock data only)

- ✓ Simple authentication (no OAuth/2FA for MVP)
- ✓ School project → functionality over perfection

QUESTIONS BEFORE WE START?

- Should we include a **dark mode** toggle?
- Want **email verification** for registration?
- Should matches have **time scheduling**?
- Any **specific games** to include in sample data?

Otherwise, let's proceed to **PHASE 1**! 🚀